Scoring universal classification schemes with hierarchical-clustering objectives

David Awad

Abstract

Library classification schemes are compared mostly by description: their history, notation, coverage and licence. Quantitative comparisons exist, but they measure retrieval effectiveness on a test collection (Cleverdon 1960; Carpenter, Jones, and Oppenheim 1978), the scatter of a topic or class across schemes (O'Neill, Dillon, and Vizine-Goetz 1987; Connaway and Sievert 1996), or coverage of a reference map of knowledge (Zins and Santos 2011). We found no study that scores the hierarchy of several universal or near-universal schemes on one structural objective over the same set of works. We propose such a comparison, built on objectives from the theory of hierarchical clustering, and we test how far it can be trusted. A scheme, restricted to the classes that a corpus actually uses, is a rooted tree over the works. We score that tree with Dasgupta's cost (Dasgupta 2016) and Moseley and Wang's revenue (Moseley and Wang 2023) on a fixed k-nearest-neighbour cosine graph of text embeddings, and normalize revenue against an average-linkage tree. We call this ratio skill. The corpus is 10,000 arXiv papers with CC0 metadata, a holdout of 500 Open Library editions with librarian-assigned LC call numbers, and a sample of 3,000 books that also carry Dewey numbers. We keep source-assigned ("gold") placements apart from placements made by one lexical classifier. On gold labels, arXiv's own taxonomy reaches skill 0.786 on arXiv papers and LCC reaches 0.419 on the book holdout. With the classifier's placements, the same two schemes fall to 0.128 and 0.005. The findings are mostly cautions. (1) Coverage drives most classifier-placed rankings: across 13 schemes, skill ranges from 0.059 (msc) to 0.423 (openalex-topics), its Spearman correlation with coverage is 0.76, and coverage explains 59% of the rank variance of cost. (2) The metric agrees only weakly with a blind LLM judge of librarian-style decisions (Cohen's $\kappa = 0.27$ on 953 pairs). A false-join term raises κ to 0.48, but it does not raise agreement with citation or subject-heading signals. (3) Rankings of the schemes on the papers are stable across embedding model, k and graph threshold within one classifier (minimum Spearman ρ 0.90) and not across classifier tiers (minimum 0.04), and objectives disagree about FDC against LCC. (4) On 3,000 books, Dewey sections against LCC is a tie that depends on the setting. (5) A learned scheme's lead over every human scheme survives rescoring on two independent embedding models, is partly home advantage, and on books does not survive the false-join correction. (6) Under a coverage-floor rule, the LCC outline with LCSH vocabulary is the default among open schemes in 27 of 27 book settings, partly through its vocabulary. The measurement is cheap and its inputs are pinned by hash. Its weak points are the classifier and, for now, the single annotator behind the validity check. We also report a classifier fix found by the benchmark: the largest single-class capture in ILC2 fell from 73.7% to 4.2% of placed papers.

1. Introduction

A universal classification scheme gives every work one address in a single tree. Library of Congress Classification (LCC), Dewey (DDC), UDC, the Basic Classification (BK), the Integrative Levels Classification (ILC) and others each claim to arrange all of knowledge. A practitioner choosing among them, or a researcher designing a new one, needs to ask which arrangement keeps related works close together, and at what navigation cost. The literature answers this mostly qualitatively, by notation, facet structure, history and institutional adoption (Nocetti 1978; Zins and Santos 2011; Oh 2023; Wu 2023). Quantitative comparisons have measured retrieval performance (Cleverdon 1960; Keen 1973; Carpenter, Jones, and Oppenheim 1978) or the scatter of a subject across schemes (O'Neill, Dillon, and Vizine-Goetz 1987; Connaway and Sievert 1996), not the structure of the hierarchy

itself (Section 2). A structured search of JASIST, Knowledge Organization, Cataloging & Classification Quarterly, ISKO and SIG/CR proceedings, JCDL and TPDL (research/reports/8-prior-work-search.md) found no study that scores several universal schemes on a common hierarchical objective over the same corpus, and no use of Dasgupta cost or Moseley–Wang revenue on a human-built scheme. The search covered titles and abstracts, not full texts (see Limitations), so this is absence of evidence, not proof.

Three developments make the question more pressing.

1. The main lookup service is gone. OCLC Classify returned the most common DDC and LCC numbers for a work from WorldCat holdings. OCLC discontinued it on 31 January 2024 (OCLC 2024). No free service replaced it. Its successor endpoint is in the WorldCat Metadata API, which is open only to subscribers.
2. Automatic placement has not closed the gap. Annif, the most mature open subject-indexing toolkit, is trained separately for each vocabulary (National Library of Finland 2026). On more than 230,000 Swedish union-catalogue records its best (ensemble) configuration reaches 66.82% accuracy at three-digit DDC (Golub et al. 2024). Recent work that uses large language models (LLMs) for LCC and DDC assignment reports weakness with more granular classes (Song et al. 2026) and a lack of specificity (Holstrom 2025), and prompted LLMs fall well short of fine-tuned smaller models on DDC classes for research data (Shahi, Shigapov, and Hummel 2025). The Library of Congress's own experiments predicted subject and genre with under 50% accuracy, short of a 95% automation threshold (Saccucci and Potter 2025). Placement is getting cheaper, so the open question shifts towards *which scheme* to place into.
3. Licences constrain open use. DDC is copyrighted by OCLC (OCLC, n.d.). The full UDC is licensed (UDC Consortium, n.d.). Openly licensed universal schemes are shallow, such as BK (BARTOC, n.d.) and FDC (Ockerbloom, n.d.), or their licence terms are unsettled, as for ILC2 (ISKO Italia, n.d.). Data-driven vocabularies such as OpenAlex Topics are open but cover research literature only (OpenAlex 2026b, 2026a). A quantitative comparison helps decide whether an open scheme is good enough to adopt.

Contributions. (i) A scheme-agnostic scoring method. It induces a tree from a scheme and a set of placements, and scores the tree with Dasgupta cost and Moseley–Wang revenue on a fixed similarity graph. The score is normalized against an average-linkage reference tree on the same graph, and coverage, structure and navigation statistics are reported with it (Section 3). (ii) A pinned, CC0-metadata corpus of 10,000 arXiv papers and a 500-edition LC holdout (Section 4). (iii) Results for 13 schemes. Gold placements and classifier placements are kept apart, and the classifier bias is calibrated where both exist (Section 5.1 to 5.4). (iv) A methods note on a classifier defect that the benchmark exposed, with its before-and-after effect (Section 5.5). (v) A validity check of the cost against a blind judge, a false-join correction, and non-LLM checks (Sections 5.6 and 5.7). (vi) Stability tests: a coverage control, a second classifier tier, and a sweep over embedding model, k and graph threshold on the papers (Sections 5.8 and 5.9), and the same sweep for the claims of a book-structure proposal on 3,000 books with Dewey numbers (Sections 5.10 and 5.11).

2. Related work

Comparing classification schemes. The Cranfield project compared the Universal Decimal Classification, an alphabetical subject catalogue, a faceted classification and Uniterm coordinate indexing, applied to 18,000 aeronautics documents (Cleverdon 1960; Jahoda 1961). Keen compared five index languages built for a test collection on retrieval effectiveness (Keen 1973), and Carpenter et al. compared five patent classifications on recall and precision for the same questions (Carpenter, Jones, and Oppenheim 1978). O'Neill et al. measured how the classes of LCC and DDC disperse into each other on MARC records (O'Neill, Dillon, and Vizine-Goetz 1987), and Connaway and Sievert compared the scatter of one subject across LCC, DDC and the NLM Classification (Connaway and Sievert 1996). Zins and Santos mapped the top two levels of LCC, DDC and UDC onto one reference model of knowledge (Zins and Santos 2011). In bibliometrics, research classifications are scored on citation-based criteria: algorithmic journal maps against the ISI Subject Categories (Rafols and Leydesdorff 2009), the Web of Science and Scopus journal classifications against each other (Q. Wang and Waltman 2016), and taxonomies built from different citation relations against one gold standard (Klavans and Boyack 2017). These studies score schemes by retrieval, by the scatter of a mapping or by citation coherence. None scores the tree itself. For a single vocabulary, corpus-weighted pruning measures how much of a subject tree a collection needs (Julien et al. 2013), checking

functions detect quality defects in SKOS vocabularies (Mader, Haslhofer, and Isaac 2012), and structural analysis is one of the established ways to evaluate a thesaurus (Owens and Cochrane 2004).

Objectives for hierarchical clustering. Dasgupta defined the cost of a tree T over items with pairwise similarities w_{ij} as $\sum_{i<j} w_{ij} \, |\text{leaves}(T[i \vee j])|$, where $i \vee j$ is the lowest common ancestor (LCA) of i and j (Dasgupta 2016). Similar pairs should be separated as deep in the tree as possible. A flat tree has the maximum cost. Moseley and Wang studied the dual, revenue, $\sum w_{ij}(n - |\text{leaves}(T[i \vee j])|)$, so that cost plus revenue equals $n \sum w_{ij}$. They showed that average linkage is a 1/3-approximation to the maximum revenue (Moseley and Wang 2017, 2023). Optimizing Dasgupta cost exactly is NP-hard. Approximation algorithms use sparsest cut and spreading metrics (Charikar and Chatziafratis 2017; Roy and Pokutta 2016). Cohen-Addad et al. give an axiomatic family of admissible objectives and show when average linkage and its variants recover planted hierarchies (Cohen-Addad et al. 2019). Hyperbolic embeddings give a differentiable relaxation (Chami et al. 2020), and structural constraints can be added to the search (Chatziafratis, Niazadeh, and Charikar 2018). scikit-network ships a Dasgupta cost and score for binary dendrograms (scikit-network developers 2026). This line of work treats the objective as something to *optimize*. We use it only to *evaluate* trees that people built.

No objective-free optimum. Kleinberg's impossibility theorem says no clustering function satisfies scale invariance, richness and consistency together (Kleinberg 2002). Carlsson and Mémoli characterize single linkage axiomatically among hierarchical methods (Carlsson and Mémoli 2010). One consequence for us: a score is always relative to the similarity function. We therefore fix and publish the graph and treat it as part of the measurement.

Navigation. Optimal search trees and Huffman coding minimize expected depth under an access distribution (Huffman 1952; Knuth 1971). The hierarchical-identity model of Watts, Dodds and Newman makes greedy navigation on a hierarchy a measurable success criterion (Watts, Dodds, and Newman 2002). Basic-level categories explain why a natural middle level carries most human use (Rosch et al. 1976). User studies of menu hierarchies disagree on the best shape. Speed and accuracy improved with breadth in one study (Snowberry, Parkinson, and Sisson 1983), and task time rose with depth for web links (Zaphiris 2000), but others found an optimum at moderate breadth (Miller 1981; Larson and Czerwinski 1998). Landauer and Nachbar fit the time per menu screen as $k + c \log b$ for b choices (Landauer and Nachbar 1985), the form of our `nav_log` model (Section 3.5). For a two-level card file, the optimum number of records per guide is proportional to the square root of the file size (Lipetz and Song 1970). We report simple, uniform-use navigation costs. We fit no user model.

Data-driven and LLM-built taxonomies. Waltman and van Eck cluster publications on direct citation at several resolutions (Waltman and Eck 2012), using the Leiden algorithm (Traag, Waltman, and Eck 2019). OpenAlex Topics come from citation clustering and are named by an LLM, and their upper levels are mapped from Scopus ASJC categories (Priem, Piwowar, and Orr 2022; OpenAlex 2026b). TaxoGen and TaxoExpan build and extend topic taxonomies from text (Zhang et al. 2018; Shen et al. 2020). LLM pipelines generate layers or labels: Chain-of-Layer, TnT-LLM and TaxoAdapt (Zeng et al. 2024; Wan et al. 2024; Kargupta et al. 2025). None of these optimizes a global objective. Taxonomy-quality work catalogues attributes and measures (Unterkalmsteiner and Abdeen 2023). It includes a reference-free robustness score that correlates semantic and taxonomic similarity (Wullschleger et al. 2026), which is closest in spirit to ours.

Automatic subject classification. Annif (National Library of Finland 2026), its SemEval-2025 Task 5 entry (Suominen, Inkinen, and Lehtinen 2025) and the task overview (D'Souza et al. 2025) cover subject indexing with thesauri such as GND. They report results per vocabulary, not across schemes. An open LCC classifier covers 15 of the 21 top-level classes (ahmad-PH 2026). We found no public leaderboard that compares classification schemes, as opposed to classifiers.

3. Method

3.1 Similarity graph

Each item's text is `title + ". " + abstract` (titles only for the book holdout, which has no descriptions). It is embedded with sentence-transformers/all-MiniLM-L6-v2 at a pinned ONNX revision (Reimers and Gurevych 2019; W. Wang et al. 2020; sentence-transformers, n.d.; Qdrant, n.d.), and vectors are L2-normalized. Raw cosine similarities between unrelated texts are not centred on zero, so we subtract the exact mean cosine τ over all

unordered pairs: $\tau = 0.109$ on the arXiv corpus and 0.173 on the holdout. Each item keeps its $k = 20$ most similar items. The weight of an edge is $w_{ij} = \max(0, \cos_{ij} - \tau)$. The graph is symmetrized by taking the maximum, and zero-weight edges are dropped. That leaves 148,111 edges on the corpus and 7,297 on the holdout. Every scheme and the reference tree are scored on this one graph, so all comparisons on a corpus are paired. We used a different embedding model for similarity than the classifier's optional embedding tier (BGE-small, used only in the semantic-tier comparison of Section 5.8). This follows the benchmark design, which aims to avoid a circular advantage for schemes whose labels embed well (research/benchmark/DESIGN.md §2.2).

3.2 From a scheme and placements to a tree

A scheme is a set of classes, each with at most one parent, so it is a tree or a forest. For polyhierarchies such as EuroVoc and the UNESCO Thesaurus, the port has already chosen one primary parent per concept. Given one placement per item, we keep the used classes and all their ancestors. Forest roots hang under a virtual root. Each item becomes a leaf under its class node, including when that class is an internal node. An item with no placement becomes a leaf directly under the root, so every pair involving it pays the full corpus size. Coverage is therefore penalized *inside* the cost, and we also report it separately. All reported runs use one class per item, the first listed code for gold labels and the classifier's top choice otherwise. The harness also supports fractional (multi-copy) placement, which we leave for later work.

3.3 Cost, revenue and skill

Let n be the number of items (placed or not), E the edges of the graph, $W = \sum_{(i,j)\in E} w_{ij}$, and $N(v)$ the number of items in the subtree of node v. Then

$$\text{cost}(T) = \sum_{(i,j)\in E} w_{ij}\, N(\text{lca}_T(i,j)), \qquad \text{cost}_{\text{norm}} = \frac{\text{cost}(T)}{nW}, \qquad \text{revenue}_{\text{norm}} = 1 - \text{cost}_{\text{norm}}.$$

A tree with every item under the root has $\text{cost}_{\text{norm}} = 1$. By the Moseley–Wang identity, $nW - \text{cost}(T)$ is the revenue, so $\text{revenue}_{\text{norm}}$ is revenue over its trivial upper bound. The sum runs over graph edges only. This is a kNN-sparsified Dasgupta cost: the pairs that matter are those the embedding calls close.

Revenue depends on the corpus and the graph, so we normalize it against a reference tree T_{ref}. That tree is a deterministic average-linkage (UPGMA) dendrogram on the same graph, with missing edges treated as similarity 0:

$$\text{skill}(T) = \frac{1 - \text{cost}_{\text{norm}}(T)}{1 - \text{cost}_{\text{norm}}(T_{\text{ref}})}.$$

Skill is 0 for the flat tree and 1 for a tree as good as average linkage. Average linkage is a 1/3-approximation of maximum revenue (Moseley and Wang 2023), so the reference is an achievable tree, not an optimum. Skill may exceed 1. The reference reaches $\text{cost}_{\text{norm}} = 0.152$ on the corpus and 0.301 on the holdout.

The reference is a binary tree over individual items. A scheme's tree stops at classes, and all items inside one class share an LCA. This tie is intended: it models scanning the class shelf. It also means that no scheme whose classes hold more than one item can reach skill 1 easily. Skill compares a scheme with a data-fitted tree. It is not a pass mark.

3.4 Gold and shelfmark-placed subsets

A score depends on the scheme *and* on how works were placed into it. We keep two subsets apart.

- Gold (track A). Labels supplied by the source. These are arXiv primary categories, author-supplied MSC and JEL codes on the arXiv corpus, and LC call numbers on the book holdout. Call numbers are resolved to the deepest covering class of the bundled LCC outline (`lcc`) and of the complete LCC schedules (`lcc-full`). The resolver tries the full call number, then the class number without cutters, then the class letters.

- Placed (track P). Labels assigned by shelfmark's lexical classifier (`profile:lexical@2`). Each scheme gets an isolated store with no hand-written rules. Placements below the classifier's 0.8 confidence target are kept and flagged. Fallback placements, where no tier produced a candidate, count as unplaced.

The two subsets are never ranked against each other. Where both exist for the same scheme and items (arXiv, MSC and JEL on the corpus; LCC and `lcc-full` on the holdout), their difference measures the classifier's effect (Section 5.3).

3.5 Structure, coverage and navigation

For each run we report:

- Coverage: the share of items with a placement.
- Depth: the mean, median, 95th percentile and maximum item depth, in edges from the root to the item leaf. The root is depth 0, and an item placed in a top-level class has depth 2.
- Branching: the fan-out of used internal nodes. Item leaves count as children, so unplaced items inflate root fan-out.
- Balance: the effective number of classes $\exp(H)$ and the Gini coefficient. Both use the distribution $p_v \propto N(v)$ over used class nodes at *all* levels. An item therefore counts once at each ancestor class, so these are multi-level summaries, not leaf-partition statistics.
- Expected navigation cost under uniform use, with two menu models. For an item i with ancestors v of fan-out $b(v) > 1$: $\mathrm{nav}_{\log}(a) = \frac{1}{n}\sum_i \sum_v (a + \log_2 b(v))$, a Hick–Hyman-style decision cost with per-level overhead $a \in \{0, 1, 2\}$. The second model is $\mathrm{nav}_{\mathrm{scan}} = \frac{1}{n}\sum_i [\sum_v (b(v) + 1)/2 + (n_c + 1)/2]$: a linear menu scan plus a scan of the n_c items under the item's deepest class. These parameters are assumptions, not fitted values. We report the sweep.

3.6 Implementation

The harness is a Rust crate, `shelfmark-bench`. It walks parent pointers to find LCAs, which the shallow class trees make cheap, and computes the average-linkage reference once per graph with a size-weighted (Lance–Williams) update and deterministic tie-breaking. Unit tests check a hand-computed cost, the identity cost + revenue $= nW$, that a root-only tree has $\mathrm{cost}_{\mathrm{norm}} = 1$, and that the embedder, the graph and the reference are deterministic. Each result records the SHA-256 of the graph, the induced tree and the scheme file.

3.7 Validity check and false-join term

Dasgupta cost charges a scheme for separating similar items. It charges nothing for joining dissimilar items, except through the graph edges the join happens to contain. To ask whether the cost tracks what a librarian would call a good decision, we drew 953 pairs from eight runs in three strata (similar pairs the scheme splits at depth one or less, dissimilar pairs it joins deep, and random placed pairs) and had one LLM judge label each blind under a rubric fixed before judging (`docs/validity`). The metric flags a pair when its cost contribution exceeds the run's mean contribution per graph edge. Agreement is Cohen's κ (Cohen 1960), with percentile bootstrap intervals that resample pairs within strata (Efron 1979). The correction adds a false-join term. For a seeded sample of non-edge pairs whose cosine is below τ, weighted by dissimilarity $d = \tau - \cos$, the specificity form is $\mathrm{FJs} = \sum d \ln(n/N(\mathrm{lca}))/(\ln n \sum d)$, so that deep joins of dissimilar items cost more than shallow ones. The corrected objective is $J' = C + \beta\,\mathrm{FJs} + \lambda N$ with $\beta = 1$, $\lambda = 0.002$ and N the `nav_log` cost in bits. An outside-mass form is also computed. Paired differences between runs use an item bootstrap on the same draws, and multiple comparisons are Holm-adjusted (Holm 1979).

4. Corpus

arXiv corpus. The arXiv OAI-PMH endpoint (`arXivRaw`) was harvested for records with datestamps from 2026-01-01 to 2026-10-09 in eight top-level groups. A seeded SHA-256 ranking then sampled 10,000 records, proportional

to group size with a floor of 100 per group (Table 1). arXiv applies CC0 1.0 to metadata (arXiv, n.d.b). Datestamps follow record updates, so older papers appear: 69.7% of identifiers carry a 2026 prefix. Computer science, mathematics and physics make up 86.4% of the sample. Gold labels in other schemes are sparse. 1,325 papers carry an MSC code, 317 an ACM class field and 2 a JEL code.

Table 1. Composition of the arXiv corpus by primary category.

arXiv top-level group (primary category)	papers	share
cs	3,968	39.7%
econ	185	1.9%
eess	519	5.2%
math	2,042	20.4%
physics	2,630	26.3%
q-bio	172	1.7%
q-fin	139	1.4%
stat	345	3.5%
total	10,000	100.0%

Open Library holdout. 500 Open Library edition records with a non-empty `lc_classifications` array were drawn through the Open Library Search and editions APIs (Open Library, n.d.a). They hold titles only, with no description: 0 rows have text beyond the title, and titles average 3.5 words. The holdout is skewed: 382 records (76.4%) have a first call number in class P (language and literature). The other 14 letters seen are B 32, H 19, Q 8, D 5, M 4, R 3, U 2, N 2, C 2, J 1, G 1, E 1, L 1. 37 records hold only an empty string in the array. Open Library asks contributors to share under CC0 1.0 (Open Library, n.d.b). Both files are pinned by SHA-256 and by ID lists.

Book sample with Dewey numbers. A third file holds 3,000 Open Library editions that each carry an LC call number, a Dewey number and subjects (320 also have a description). The 21 LCC main classes have equal quotas, and no edition shares a work with the 500-book holdout. The graph text is the description followed by up to 30 subject headings, so it is mostly Library of Congress subject headings (`bench/corpus/holdout-books.jsonl`). The sample's τ is 0.157. Dewey numbers are used as notation only: the tree is the digit prefixes, and no Dewey caption, schedule or table is read (Section 7). A work-level lookup finds a Dewey number for many more of the 500-book holdout's works than its edition records carry (research/reports/9, Section 3).

Schemes. Table 2 lists the 13 schemes as built for the run. Five are general or universal: LCC (outline and complete schedules), FDC, BK and ILC2. The rest are thesauri or vocabularies (UNESCO Thesaurus, EuroVoc, OpenAlex Topics) or domain schemes (arXiv, MSC, JEL, ACM CCS, IPC). Each scheme file's SHA-256 is recorded, and the r2 run reused byte-identical files.

Table 2. Schemes, class counts and licence as recorded by the build.

scheme	version	classes	licence as recorded by the build
`acm-ccs`	2012	2,113	ACM, Inc.: educational and research use; not redistributable, so not bundled
`arxiv`	2024-04-01	192	see scheme README
`bk`	2026-10-03	2,140	CC0 1.0

scheme	version	classes	licence as recorded by the build
`eurovoc`	4.24	7,663	EU reuse terms stated inconsistently (Commission reuse notice in dataset metadata, CC0 in a third-party registry); not bundled, built from the Publications Office download
`fdc`	0.06	1,097	CC0 1.0
`ilc2`	2019-10-17	10,117	Unsettled: the publisher's SKOS file says CC BY 3.0, BARTOC lists CC BY-SA 3.0, no iskoi.org page states a license; not bundled
`ipc`	2026.01	80,145	CC BY 4.0
`jel`	2026	1,015	see scheme README
`lcc`	2024	7,392	see scheme README
`lcc-full`	2024	416,186	US government work, no US copyright (17 U.S.C. 105)
`msc`	2020	6,603	see scheme README
`openalex-topics`	2026-10-09	4,798	CC0 1.0
`unesco-thesaurus`	2026-08-31	4,595	CC BY-SA 3.0 IGO (https://creativecommons.org/licenses/by-sa/3.0/igo/)

The scheme READMEs give the licences for the rows marked "see scheme README". arXiv's taxonomy comes from arxiv-base, under MIT (arXiv, n.d.a). MSC2020 is CC BY-NC-SA 4.0 (Mathematical Reviews and zbMATH Open, n.d.). JEL publishes no open licence (American Economic Association, n.d.). The LCC outline is built from Library of Congress publications (Library of Congress 2026b, 2026a; Legal Information Institute, n.d.). Sources for the others: IPC (WIPO, n.d.a, n.d.b), ACM CCS (ACM, n.d.), EuroVoc (Publications Office of the European Union, n.d.), UNESCO Thesaurus (UNESCO, n.d.).

5. Results

Every table in this section is generated from the published JSON (Section 7). Rows are sorted by $\text{cost}_{\text{norm}}$, with the best first.

5.1 Gold placements

Table 3. Gold placements on the arXiv corpus (bench/results/2026-10-10; `msc` rescored on the fixed corpus in bench/results/2026-10-10-r3).

scheme	version	coverage	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log (a=1)
arxiv	2024-04-01	100.0%	0.333	0.667	0.786	36.5	3.90	4	797	0.795	18.3
msc	2020	13.2%	0.984	0.016	0.018	461.5	1.39	4	25	0.599	15.2
jel	2026	0.02%	1.000	0.000	0.000	6.0	1.00	4	1	0.000	14.3

Table 4. Gold placements on the book holdout. LC call numbers are resolved to the deepest covering class.

scheme	version	coverage	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log (a=1)
lcc-full	2024	92.2%	0.706	0.294	0.421	85.2	5.19	10	68	0.718	19.7
lcc	2024	92.2%	0.708	0.292	0.419	35.0	4.20	9	89	0.791	18.6

arXiv's own taxonomy, applied to arXiv papers by their authors, reaches skill 0.786 with full coverage. This is the highest score in the study. We read it as an upper reference for what a well-fitted scheme with correct placements achieves on this graph, not as evidence that arXiv's taxonomy is a good *universal* scheme. The arXiv categories and the abstracts come from the same authors, and the corpus was sampled by arXiv group.

On the books, LCC call numbers resolved into the outline reach skill 0.419 and into the complete schedules 0.421, at the same coverage (92.2%). The complete schedules use more effective classes (85 vs 35) and are deeper (5.19 vs 4.20 mean depth). Yet cost barely moves (0.706 vs 0.708). With 500 short titles and a holdout that is three quarters class P, the title-embedding graph does not reward the finer LCC distinctions. This does not mean the distinctions are useless; it is a limit of what the graph can see.

MSC and JEL gold results are uninformative. With 13.2% and 0.02% coverage, almost every pair involves an item under the root. ACM CCS has no gold run at all. The corpus's `acm_ccs` field holds the ACM classes submitted to arXiv, and no value in it is a CCS 2012 concept path. The run log records: "the corpus acm_ccs field (690 values on 317 papers) holds ACM CCS 1998 codes such as `I.2.6`, as submitted to arXiv; the scheme is CCS 2012, whose codes are paths of 8-digit concept IDs, and no CCS 1998 to 2012 crosswalk is used".

5.2 shelfmark-placed results

Table 5. Placed by shelfmark (`profile:lexical@2`) on the arXiv corpus (bench/results/2026-10-10-r2).

scheme	version	coverage	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log (a=1)
openalex-topics	2026-10-09	97.3%	0.641	0.359	0.423	77.6	4.45	5	554	0.892	26.9
unesco-thesaurus	2026-08-31	97.3%	0.721	0.279	0.329	146.7	4.90	9	535	0.852	26.7
ipc	2026.01	98.0%	0.769	0.231	0.272	317.8	6.31	10	794	0.838	27.3
ilc2	2019-10-17	100.0%	0.819	0.181	0.214	295.1	5.09	13	417	0.824	23.4
lcc-full	2024	100.0%	0.823	0.177	0.209	657.2	6.06	11	121	0.803	22.9
acm-ccs	2012	74.5%	0.831	0.169	0.200	84.9	3.61	7	877	0.861	23.7
fdc	0.06	77.0%	0.834	0.166	0.196	54.5	3.21	7	1103	0.836	25.6
eurovoc	4.24	97.4%	0.848	0.152	0.179	269.3	4.98	8	350	0.821	24.8
lcc	2024	87.7%	0.849	0.151	0.178	129.0	4.08	10	661	0.837	25.4

scheme	version	coverage	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log (a=1)
arxiv	2024-04-01	42.6%	0.891	0.109	0.128	14.5	2.21	4	1665	0.856	20.3
bk	2026-10-03	48.1%	0.902	0.098	0.116	75.0	3.15	8	788	0.841	20.9
jel	2026	47.9%	0.903	0.097	0.115	28.9	2.27	4	972	0.806	19.4
msc	2020	39.2%	0.950	0.050	0.059	114.4	2.00	4	380	0.800	17.8

Table 6. Placed by shelfmark on the book holdout (general schemes only).

scheme	version	coverage	cost_norm	revenue_norm	skill	eff. classes	mean depth	max depth	max fan-out	Gini	nav_log (a=1)
lcc-full	2024	56.6%	0.925	0.075	0.107	340.8	4.14	11	13	0.492	14.0
ilc2	2019-10-17	45.6%	0.954	0.046	0.065	188.1	2.94	11	25	0.527	12.5
unesco-thesaurus	2026-08-31	15.8%	0.989	0.011	0.016	93.8	1.65	7	8	0.439	10.8
lcc	2024	8.4%	0.996	0.004	0.005	63.0	1.29	8	6	0.329	10.2
fdc	0.06	5.2%	0.998	0.002	0.003	31.1	1.14	7	4	0.314	10.1
bk	2026-10-03	4.4%	0.999	0.001	0.002	30.8	1.23	7	7	0.439	10.1

Table 7. Placement statistics for the classifier runs. “Flagged” placements are below the 0.8 confidence target but kept. “Top matched term” is the vocabulary term that sent items to the top class.

scheme	corpus	rows	placed	flagged (<0.8)	unplaced	classes used	top class	top share	top matched term
acm-ccs	corpus	10,000	7,445	6,854	2,555	405	10002944.10011123.10010674	11.8%	“performance”
arxiv	corpus	10,000	4,260	3,325	5,740	72	cs.PF	39.1%	“performance”
bk	corpus	10,000	4,806	3,980	5,194	213	01.00	16.4%	“general”
eurovoc	corpus	10,000	9,737	9,582	263	802	C2228	3.6%	“paper”
fdc	corpus	10,000	7,696	7,023	2,304	231	676	14.3%	“paper”
ilc2	corpus	10,000	9,999	9,999	1	1,103	edqam	4.2%	“s”
ipc	corpus	10,000	9,799	9,725	201	853	G09B23/00	8.1%	“models”
jel	corpus	10,000	4,786	3,854	5,214	62	M53	20.3%	“training”
lcc	corpus	10,000	8,765	8,320	1,235	379	QA299.6-433	7.5%	“analysis”
lcc-full	corpus	10,000	9,999	9,999	1	2,421	QC243.3.E5	1.2%	“energy”
msc	corpus	10,000	3,916	2,999	6,084	324	51-XX	9.7%	“geometry”
openalex-topics	corpus	10,000	9,732	9,616	268	1,020	F21	5.7%	“energy”
unesco-thesaurus	corpus	10,000	9,731	9,529	269	675	c155	5.5%	“time”
bk	holdout	500	22	22	478	13	71.32	31.8%	“man”
fdc	holdout	500	26	26	474	20	750	11.5%	“art”
ilc2	holdout	500	228	226	272	151	edqam	11.0%	“s”
lcc	holdout	500	42	42	458	31	QH501-531	7.1%	“life”
lcc-full	holdout	500	283	283	217	227	K23	2.1%	“s”
unesco-thesaurus	holdout	500	79	77	421	60	c155	5.1%	“time”

On the corpus, skill under classifier placement runs from 0.059 (msc) to 0.423 (openalex-topics). Second is unesco-thesaurus at 0.329. Coverage explains much of the order. Across the 13 schemes, the Spearman correlation between coverage and skill is 0.76. Among the 6 schemes with at least 95% coverage, skill still spreads from 0.179 to 0.423, so coverage does not explain everything. Within that group, OpenAlex Topics scores best. It was built from research literature, the same kind of text as the corpus (OpenAlex 2026b). EuroVoc, a policy and legal thesaurus, scores lowest. That fits the expectation that a vocabulary matched to the corpus domain organizes it better. But Table 7 shows the classifier produces many low-confidence placements: in 9 of the 13 corpus runs, over 90% of placed items are flagged. The classifier's reach is itself part of the score (Sections 5.3 and 5.8).

The holdout results are near zero for every scheme except `lcc-full` and `ilc2`. On titles of 3.5 words, the lexical classifier finds a candidate for only 4.4% (BK) to 56.6% (complete LCC) of the books.

5.3 How large is the classifier effect?

Table 8 compares gold and classifier placements for the same scheme on the same items and graph.

Table 8. Gold vs classifier placement on identical items.

scheme	corpus	gold coverage	placed coverage	gold skill	placed skill	placed − gold
`arxiv`	corpus	100.0%	42.6%	0.786	0.128	-0.658
`msc`	corpus	13.2%	39.2%	0.018	0.059	+0.041
`jel`	corpus	0.02%	47.9%	0.000	0.115	+0.115
`lcc`	holdout	92.2%	8.4%	0.419	0.005	-0.413
`lcc-full`	holdout	92.2%	56.6%	0.421	0.107	-0.314

Where gold coverage is high (arXiv, LCC), the classifier loses most of the skill. For arXiv, skill falls from 0.786 to 0.128. The classifier places only 42.6% of the papers and sends 39.1% of those to cs.PF (Performance) on the word "performance" (Table 7). For the LCC outline on books, skill falls from 0.419 to 0.005. Where gold coverage is tiny (MSC, JEL), the classifier scores *higher* than the gold labels, only because it places more items. Neither direction measures scheme quality. In this run, the classifier, not the scheme, accounts for most of the differences in classifier-placed skill. The ranking in Table 5 is therefore a ranking of *scheme as reached by this classifier*. We do not present it as a ranking of schemes.

5.4 Navigation

Table 9. Depth and navigation cost under uniform use (items in the root menu included).

scheme	labels	corpus	p50 depth	p95 depth	root fan-out	nav_log a=0	a=1	a=2	nav_scan
`arxiv`	gold	corpus	4	4	8	15.2	18.3	21.5	291
`jel`	gold	corpus	1	1	10000	13.3	14.3	15.3	5001
`msc`	gold	corpus	1	4	8738	13.9	15.2	16.5	4372
`lcc`	gold	holdout	4	6	53	14.8	18.6	22.4	69
`lcc-full`	gold	holdout	5	7	53	15.4	19.7	24.1	56
`acm-ccs`	placed	corpus	4	6	2568	20.5	23.7	27.0	1476
`arxiv`	placed	corpus	1	4	5748	18.5	20.3	22.2	3231
`bk`	placed	corpus	1	6	5199	18.3	20.9	23.5	2736
`eurovoc`	placed	corpus	5	6	284	20.2	24.8	29.4	269
`fdc`	placed	corpus	4	5	2314	22.4	25.6	28.7	1413
`ilc2`	placed	corpus	5	7	68	18.9	23.4	27.9	136
`ipc`	placed	corpus	6	9	209	22.3	27.3	32.3	322
`jel`	placed	corpus	1	4	5231	17.4	19.4	21.4	2876

scheme	labels	corpus	p50 depth	p95 depth	root fan-out	nav_log a=0	a=1	a=2	nav_scan
lcc	placed	corpus	4	6	1255	21.7	25.4	29.0	825
lcc-full	placed	corpus	6	8	22	17.5	22.9	28.2	52
msc	placed	corpus	1	4	6141	16.0	17.8	19.7	3128
openalex-topics	placed	corpus	5	5	272	22.4	26.9	31.3	311
unesco-thesaurus	placed	corpus	5	7	276	22.0	26.7	31.4	305
bk	placed	holdout	1	1	481	9.1	10.1	11.2	242
fdc	placed	holdout	1	2	482	9.0	10.1	11.2	243
ilc2	placed	holdout	1	7	300	10.3	12.5	14.6	155
lcc	placed	holdout	1	4	470	9.1	10.2	11.3	237
lcc-full	placed	holdout	5	9	236	11.3	14.0	16.7	125
unesco-thesaurus	placed	holdout	1	6	427	9.4	10.8	12.2	216

The navigation models answer a different question from cost: how much reading it takes to reach an item, not whether related items are near each other. With unplaced items in the root menu, the two models diverge. `nav_log` *drops* as coverage falls, because one large root menu costs only $\log_2 b$. `nav_scan` rises steeply, because the root must be scanned linearly. Among high-coverage runs, the complete LCC schedules have the lowest scan cost (`nav_scan` 52 on the corpus) and the most effective classes. This is the deep, narrow-shelf trade-off: more levels to descend, but short shelves at the bottom. The models' parameters are assumptions, so Table 9 shows the trade-off and does not settle it. The menu studies cited in Section 2 disagree on the best depth and breadth (Miller 1981; Snowberry, Parkinson, and Sisson 1983; Larson and Czerwinski 1998), so they do not fix the parameters either.

5.5 Methods note: the stopword fix

The first placed run (bench/results/2026-10-10, `profile:lexical@1`) showed single-class captures. The classifier's vocabulary tier matched complete class labels and terms, and some of them consist only of English function words. ILC2 has a class labelled "to", IPC a C01G catchword "TO" and the complete LCC a class `PL2813` labelled "To". Any abstract containing "to" matched them. The fix (classifier `vocab-2`, `profile:lexical@2`) drops normalized labels and terms whose tokens are all stopwords. Labels with at least one content word still match in full. We then reran only the placed part, on the same graphs, scheme files (byte-identical, checked by SHA-256) and reference costs. The gold results do not depend on the classifier and were not rerun. Every placed number in this paper comes from the rerun (bench/results/2026-10-10-r2).

Table 10. Effect of the stopword fix (before: `lexical@1`, after: `lexical@2`).

scheme	corpus	top class before	top class after	coverage	cost_norm	skill	Δ skill
acm-ccs	corpus	11.8%	11.8%	74.5% →	0.831 →	0.200 →	+0.000
		10002944.10010294410016734.5011674			0.831	0.200	
arxiv	corpus	39.1%	39.1%	42.6% →	0.891 →	0.128 →	+0.000
		cs.PF	cs.PF	42.6%	0.891	0.128	
bk	corpus	16.4%	16.4%	48.1% →	0.902 →	0.116 →	+0.000
		01.00	01.00	48.1%	0.902	0.116	
eurovoc	corpus	3.6%	3.6%	97.4% →	0.848 →	0.179 →	+0.001
		C2228	C2228	97.4%	0.848	0.179	
fdc	corpus	14.3%	14.3%	77.0% →	0.834 →	0.196 →	+0.000
		676	676	77.0%	0.834	0.196	
ilc2	corpus	73.7%	4.2%	100.0% →	0.825 →	0.206 →	+0.007
		agy993	edqam	100.0%	0.819	0.214	

scheme	corpus	top class before	top class after	coverage	cost_norm	skill	Δ skill
ipc	corpus	60.1% C01G	8.1% G09B23/00	99.4% → 98.0%	0.781 → 0.769	0.259 → 0.272	+0.014
jel	corpus	18.7% M53	20.3% M53	51.7% → 47.9%	0.902 → 0.903	0.116 → 0.115	-0.001
lcc	corpus	7.5% QA299.6-433	7.5% QA299.6-433	88.4% → 87.7%	0.848 → 0.849	0.179 → 0.178	+0.000
lcc-full	corpus	22.1% PL2813	1.2% QC243.3.E5	100.0% → 100.0%	0.800 → 0.823	0.236 → 0.209	-0.027
msc	corpus	9.7% 51-XX	9.7% 51-XX	39.2% → 39.2%	0.950 → 0.950	0.059 → 0.059	+0.000
openalex-topics	corpus	5.6% F21	5.7% F21	97.4% → 97.3%	0.641 → 0.641	0.423 → 0.423	+0.000
unesco-thesaurus	corpus	17.8% c493	5.5% c155	98.9% → 97.3%	0.732 → 0.721	0.316 → 0.329	+0.013
bk	holdout	31.8% 71.32	31.8% 71.32	4.4% → 4.4%	0.999 → 0.999	0.002 → 0.002	+0.000
fdc	holdout	11.5% 750	11.5% 750	5.2% → 5.2%	0.998 → 0.998	0.003 → 0.003	+0.000
ilc2	holdout	13.7% agqs99	11.0% edqam	56.8% → 45.6%	0.934 → 0.954	0.095 → 0.065	-0.029
lcc	holdout	7.1% QH501-531	7.1% QH501-531	8.4% → 8.4%	0.996 → 0.996	0.005 → 0.005	+0.000
lcc-full	holdout	3.9% BQ940	2.1% K23	61.0% → 56.6%	0.917 → 0.925	0.119 → 0.107	-0.011
unesco-thesaurus	holdout	5.0% c155	5.1% c155	16.0% → 15.8%	0.989 → 0.989	0.016 → 0.016	+0.000

The fix removed the captures. ILC2's largest class fell from 73.7% to 4.2% of placed papers, IPC's from 60.1% to 8.1%, complete LCC's from 22.1% to 1.2% and the UNESCO Thesaurus's from 17.8% to 5.5%. Skill moved much less. On the corpus, 6 of 13 schemes changed skill by at least 0.0005, and Kendall's τ_b between the skill values before and after is 0.95. Complete LCC *lost* skill (0.236 → 0.209) even though its placements became plausible. A plausible explanation, not tested here: the wrong placements had piled most papers that share "to" into a single class. Within such a pile, the edges the graph keeps still connect similar papers, and the pile is far smaller than the whole corpus, so its edges cost less than they would under the root. A metric can reward a degenerate placement more than a spread of honest but noisy placements. This is a known property of a cost that charges same-class pairs the class size: one large class still beats the root. It is also why we always report largest-class share and placement statistics next to skill.

5.6 Does the cost agree with a judge?

Table 11 is the pooled result of the validity check (Section 3.7). The judge labels a pair's scheme decision correct or incorrect. Over 953 pairs the metric's flag agrees with the judge at $\kappa = 0.27$ [0.22, 0.32], with agreement 0.63 and an AUC of 0.65 for the raw contribution. That is weak agreement. The judge is stable against itself ($\kappa = 0.92$ on a blind second pass of 238 pairs). Strata show where the gap is. Among similar pairs the scheme splits (stratum A), the judge calls 0.70 incorrect and the metric flags 0.84. Among dissimilar pairs the scheme joins deep (stratum B), the judge also calls 0.70 incorrect and the metric flags 0.00: those pairs have zero weight, so the cost is blind to them by construction. Shares are over a stratified sample, one third per stratum, not over all pairs. Across the eight runs, the Spearman correlation between a run's `cost_norm` and the judge's error rate on random pairs is 0.23, and over all strata 0.42, from eight points. This is descriptive only. The pair sample is small per run (120 pairs), so per-run intervals are wide.

Table 11. Agreement of the metric's flag with the judge, pooled and with the false-join variants (same 953 pairs and same labels).

variant	flagged	kappa [95% CI]	AUC [95% CI]
Dasgupta cost C	0.28	0.27 [0.22, 0.32]	0.65 [0.62, 0.67]
FJs alone	0.34	0.21 [0.15, 0.26]	0.61 [0.59, 0.64]
C + 0.5 FJs	0.60	0.49 [0.44, 0.55]	0.80 [0.77, 0.83]
C + FJs	0.62	0.48 [0.43, 0.53]	0.78 [0.75, 0.81]
C + 4 FJs	0.65	0.45 [0.40, 0.51]	0.76 [0.73, 0.79]
C + FJ (outside mass)	0.61	0.47 [0.41, 0.52]	0.78 [0.75, 0.81]

5.7 A cost that sees false joins

Adding the specificity false-join term to the cost raises agreement with the judge to $\kappa = 0.48$ [0.43, 0.53], a paired change of +0.21 [+0.15, +0.27], and the AUC from 0.65 to 0.78 (Table 11). The false-join term alone does not do it ($\kappa = 0.21$); the gain comes from the sum. Weights from 0.25 to 8 give κ between 0.45 and 0.49, so the choice of β is not critical in this range. Table 12 breaks the change down by run. The correction helps the placed runs that join dissimilar items: every placed run except the learned scheme gains, with intervals that exclude zero. It does nothing for arXiv's own taxonomy or for the learned scheme, where the judge rarely finds a false join.

Table 12. Judge agreement before and after adding the false-join term, per run.

run	kappa, C [95% CI]	kappa, C + FJs [95% CI]	change in kappa
arxiv-gold	0.24 [0.05, 0.43]	0.11 [0.05, 0.18]	-0.13 [-0.28, +0.04]
msc-gold	0.20 [0.01, 0.38]	0.33 [0.21, 0.46]	+0.13 [-0.07, +0.33]
fdc-placed	0.29 [0.21, 0.38]	0.63 [0.48, 0.78]	+0.34 [+0.19, +0.47]
ilc2-placed	0.25 [0.18, 0.33]	0.66 [0.50, 0.79]	+0.40 [+0.28, +0.50]
lcc-full-placed	0.25 [0.19, 0.32]	0.68 [0.54, 0.81]	+0.43 [+0.33, +0.52]
openalex-topics-placed	0.26 [0.16, 0.35]	0.66 [0.52, 0.79]	+0.40 [+0.28, +0.50]
unesco-thesaurus-placed	0.24 [0.17, 0.32]	0.54 [0.38, 0.69]	+0.30 [+0.16, +0.43]
learned-placed	0.33 [0.16, 0.50]	0.30 [0.18, 0.43]	-0.03 [-0.20, +0.14]

Two checks that do not use an LLM temper this. The first signal is a shared Library of Congress subject heading between books. The second is bibliographic coupling or co-citation between arXiv papers, from OpenAlex reference lists (research/reports/9, Section 4.6). Table 13 gives the agreement of the pair-level flag with each signal, and the separation each signal finds between an FDC backbone and an LCC backbone. The false-join term does *not* raise pair-level agreement with any of the three (coupling 0.13 to 0.06, co-citation 0.23 to 0.19, subject headings 0.24 to 0.21). At scheme level, rank agreement is unchanged. So the corrected metric is better supported than the original, by one annotator model, and it is not better supported by independent signals. Citation and subject-heading signals have their own limits: most 2025 to 2026 preprints in the corpus have no OpenAlex reference list, and LC assigns subject headings and call numbers in one record, so that signal is not independent of LCC.

Table 13. Pair-level agreement (κ) and scheme-level rank agreement (ρ) of the cost and the corrected cost with non-LLM signals.

external signal	kappa, C	kappa, C + FJs	scheme-level rho, C	scheme-level rho, C + FJs	FDC-backbone minus LCC-backbone separation [95% CI]
shared LCSH heading (books)	0.24	0.21	0.88	0.89	ddc-section - lcc-gold: -0.002 [-0.011, +0.007]
bibliographic coupling (papers)	0.13	0.06	0.90	0.90	fdc-arxiv-nested - lcc-arxiv-nested: +0.004 [-0.001, +0.010]
co-citation (papers)	0.23	0.19	0.87	0.87	fdc-arxiv-nested - lcc-arxiv-nested: +0.001 [-0.026, +0.025]

Part of the judge's verdict is also about the schemes. The judge's together-or-apart call agrees with whether the pair is an edge of the graph at $\kappa = 0.62$, and disagrees with whether the scheme put the pair together (κ = -0.31). The judge finds, on its sample, a false-join share of 0.24 and a false-split share of 0.27 of all sampled pairs, shares that depend on how the strata were drawn.

5.8 Coverage control and the classifier tier

A gold run's unlabelled items hang under the root and are charged the maximum, so coverage enters the cost directly. The covered-subset analysis rescored the placed runs on each run's own covered items. Coverage explains 59% of the rank variance of Dasgupta cost over the 14 placed runs on the papers (ρ = -0.77), and the ranking by cost on all items and on each run's own covered items barely agrees (Kendall $\tau = 0.10$). Low-coverage schemes move from the bottom to the middle once their unplaced papers are dropped. The sparse graph is not the cause: on random subsets of 1,000 to 5,000 papers, the kNN-sparse cost and a dense cost rank the placed runs with Spearman $\rho \geq 0.92$.

A second classifier tier tests the same point from the other side. The shortlist profile adds a BGE-small embedding shortlist after the lexical tiers and files every item, so coverage is 1.000 for all 13 schemes. The rerank profile is unavailable in this build and reproduces the shortlist placements. Table 14 compares the tiers on the same graph and schemes. Skill rises for 12 of 13 schemes, by much more where the lexical tier placed few papers. arXiv's own taxonomy moves from 0.128 to 0.502 and now leads, with openalex-topics second (0.465), and the Spearman correlation between the two tiers' skill values is 0.14. The ranking of schemes under a classifier is therefore a property of the classifier-scheme pair. The run is marked pending in `bench/results/README.md`: the numbers here are read from the committed JSON of that directory and should be treated as preliminary, and the tier that places by embedding uses a different model from the graph's.

Table 14. Lexical and shortlist placement on the papers (`profile:lexical@2` and `profile:shortlist@2`; bench/results/2026-10-10-semantic).

scheme	coverage lexical	coverage shortlist	skill lexical	skill shortlist	change in skill
acm-ccs	0.745	1.000	0.200	0.278	+0.078
arxiv	0.426	1.000	0.128	0.502	+0.373
bk	0.481	1.000	0.116	0.360	+0.244
eurovoc	0.974	1.000	0.179	0.175	-0.004
fdc	0.770	1.000	0.196	0.258	+0.062

scheme	coverage lexical	coverage shortlist	skill lexical	skill shortlist	change in skill
ilc2	1.000	1.000	0.214	0.244	+0.031
ipc	0.980	1.000	0.272	0.296	+0.023
jel	0.479	1.000	0.115	0.303	+0.189
lcc	0.876	1.000	0.178	0.212	+0.033
lcc-full	1.000	1.000	0.209	0.217	+0.008
msc	0.392	1.000	0.059	0.230	+0.171
openalex-topics	0.973	1.000	0.423	0.465	+0.042
unesco-thesaurus	0.973	1.000	0.329	0.338	+0.009

5.9 Rank stability on the papers

The stability sweep rescored the 13 schemes on the papers under 54 controlled settings: three embedding models, k of 10, 20 and 50, three graph thresholds, and the lexical and shortlist tiers. A ranking claim is called robust when the minimum Spearman ρ over every pair of settings is at least 0.8 (the sweep's own rule). Table 15 gives the minimum and median ρ between settings that use the same classifier tier and between settings that do not.

Table 15. Rank agreement of the 13 schemes between controlled settings, by objective, and the number of settings in which FDC ranks above LCC.

objective	min rho, same tier	median rho, same tier	min rho, across tiers	median rho, across tiers	FDC above LCC (settings)
Dasgupta cost	0.896	0.978	0.038	0.154	0 of 54
Moseley-Wang revenue	0.896	0.978	0.038	0.154	0 of 54
skill	0.896	0.978	0.038	0.154	0 of 54
triplet accuracy	0.808	0.984	-0.280	-0.104	27 of 54
shelf-order locality	0.780	0.962	0.549	0.736	0 of 54

Within a tier, rankings are stable: cost, revenue and skill have a minimum ρ of 0.90 over the embedding, k and threshold sweep. Across tiers they are not: the minimum is 0.04 and the median 0.15. The leader under cost is openalex-topics under the lexical tier and arxiv under the shortlist tier. Objectives also disagree. FDC ranks above LCC by cost, revenue, skill and locality in 0 settings. By triplet accuracy it ranks above LCC in 27 of 54, and the triplet ranking can be reversed between settings (minimum ρ -0.28). A claim that one scheme beats another on these papers therefore depends on the classifier and on the objective. The graph settings matter much less.

5.10 Books with Dewey numbers: Dewey against LCC

On the 3,000 books every item has both a Dewey number and an LC call number from librarians, so the Dewey and LCC runs are gold on the same items and coverage is 1.000 for both. Table 16 gives the runs on the MiniLM $k = 20$ graph. Table 17 gives the comparisons that a book-structure proposal drew from them, rechecked in each of 27 settings (three embedding models, three values of k and three graph thresholds, with 1,000 paired item-bootstrap draws each).

Table 16. Books, MiniLM, $k = 20$, published graph. C is `cost_norm`, FJs enters at $\beta = 1$, J' adds $0.002N$.

run	structure	coverage	C	C + FJs	nav bits	J'	skill
ddc-gold	Dewey, librarian number, full depth	1.000	0.493	0.520	19.3	0.558	0.658
ddc-section	Dewey, cut to section (3 digits)	1.000	0.494	0.521	17.1	0.555	0.657
fdc-d3	FDC from Dewey, depth 3	1.000	0.496	0.523	17.7	0.559	0.654
fdc-d2	FDC from Dewey, depth 2	1.000	0.502	0.528	16.1	0.561	0.647
lcc-gold	LCC, librarian call number (outline)	1.000	0.523	0.531	16.2	0.563	0.619
lcc-full-gold	LCC, librarian call number (complete schedules)	1.000	0.522	0.530	16.8	0.563	0.620
learned-own	learned scheme, own insertion	1.000	0.339	0.391	29.7	0.450	0.858
fdc-placed	FDC, lexical classifier	0.839	0.654	0.680	23.4	0.727	0.450
lcc-lcsh-placed	LCC outline with LCSH vocabulary, lexical classifier	0.985	0.674	0.689	18.6	0.726	0.423
lcc-placed	LCC outline, lexical classifier	0.915	0.733	0.750	20.6	0.791	0.346
unesco-thesaurus-placed	UNESCO Thesaurus, lexical classifier	0.925	0.629	0.656	22.2	0.700	0.482

Table 17. Pair claims across 27 settings. "Claimed" is the sign the proposal reported. "Sign matches" counts settings where the difference has that sign. "Significant" counts settings whose 95% interval excludes zero, for and against the claim. A tie claim fails when any interval excludes zero.

comparison (a vs b)	objective	claimed	sign matches	significant for / against	range of a - b	verdict
Dewey sections vs LCC, librarian numbers	C	a < b	13 of 27	10 / 9	[-0.0753, +0.0750]	not stable
Dewey sections vs LCC, librarian numbers	J'	tie	8 of 27	- / 19	[-0.0564, +0.0948]	not stable
Dewey sections vs FDC depth 3	C	a < b	27 of 27	27 / 0	[-0.0044, -0.0010]	stable
FDC depth 3 vs depth 2	J'	a < b	27 of 27	24 / 0	[-0.0085, -0.0001]	stable
FDC depth 3 vs full FDC number	J'	a < b	15 of 27	9 / 1	[-0.0005, +0.0004]	not stable
Full Dewey number vs section	J'	a > b	27 of 27	27 / 0	[+0.0013, +0.0036]	stable
Learned scheme vs Dewey sections	J'	a < b	19 of 27	13 / 6	[-0.1520, +0.0917]	not stable
Placed: FDC vs LCC outline	J'	a < b	27 of 27	24 / 0	[-0.0897, -0.0237]	stable
Placed: FDC vs LCC outline + LCSH	J'	a < b	9 of 27	5 / 8	[-0.0434, +0.0729]	not stable

Dewey sections beat LCC on cost on the published graph for all three embedding models (-0.0292 [-0.0438, -0.0141] on MiniLM, -0.0411 on bge-small, -0.0239 on nomic, at $k = 20$). Across the 27 settings the sign matches in 13, with 10 significant for and 9 significant against, and the difference ranges from -0.0753 to +0.0750. Under the corrected objective the claim becomes a tie, and the tie holds in 8 settings with 19 significant departures. The worst-case regret that earlier favoured Dewey sections (0.060 against 0.067 for LCC over the nine published graphs) reverses over all 27 settings (0.116 against 0.067). The non-LLM signals do not separate them either: shared subject headings -0.002 [-0.011, +0.007], and for papers shelved under an FDC or an LCC backbone, coupling +0.004 [-0.001, +0.010] and co-citation +0.001 [-0.026, +0.025]. We read this as a tie that depends on the setting: Dewey and LCC are not separated by these metrics on these books. The sample is stratified equally over LCC's main classes, which favours LCC's false-join term, and its text is mostly LC subject headings.

Smaller comparisons are stable in every setting. A three-digit Dewey section beats FDC depth 3 by less than

0.0044 of cost (a caption-versus-number difference). Three levels beat two. The full Dewey number gains cost and loses on J'. Through the same classifier, FDC beats the LCC outline without subject-heading vocabulary in all 27 settings. It does not beat the bundled LCC outline with LCSH vocabulary, which places 0.99 of the books against 0.84 for FDC: that comparison turns on vocabulary and coverage, not on the scheme. On the 500-book title-only holdout no placed run reaches the coverage floor of 0.80, and the best reaches 0.57.

5.11 The learned scheme and home advantage

A scheme fitted to the objective by order-preserving dynamic programming and local search (the companion draft in `paper/paper2`) scores far better than every human scheme on its own corpus: in-sample skill 0.976 on the papers, and 0.922 on a held-out 20% split. Through the lexical classifier it falls to 0.732. On books, its cost is lower than Dewey sections' on every graph, but the margin is largest on the graph it was fitted on (Table 18). Refits on the other two models score best on their own graph (Table 19): on the held-out split, a fit scored on its own model's graph averages 0.926 skill against 0.904 elsewhere, and is best on its own graph for 3 of 3 models. Part of the learned scheme's lead is therefore home advantage. Across the 27 settings, the claim that it beats Dewey sections is not stable (sign matches in 19 of 27 under J', with 6 significant reversals; Table 17). It also needs the embedding model to file a new item and has an opaque notation. We use it as a reference for how low a fitted tree can go, not as a candidate scheme.

Two further checks bound the claim. Rescored on graphs from the two other embedding models at k = 10, 20 and 50, it still ranks first by skill in all 9 conditions for the held-out split, the in-sample fit, the inserted book holdout and the placed run, with paired intervals of the margin above zero (tightest: 0.155 [0.122, 0.189] on bge-small-k10). Under the corrected objective, against the best non-learned run of each universe on the false-join rescoring's graphs, it leads on the papers in 3 of 3 graphs and on the title-only holdout in 3 of 3. On the 3,000 books it leads in 6 of 9, ties in 2 and is behind in 1; under the outside-mass form it is behind in 6 of 9. Its edge survives a change of embedding and does not survive the false-join correction on books. Paper 2, Section 6.6, gives the tables.

Table 18. Learned scheme against Dewey sections on books, by graph (all $k = 20$; the scheme was fitted on MiniLM).

graph (books, k = 20)	C learned	C Dewey sections	C difference	J' difference
minilm (fit graph)	0.339	0.494	-0.155	-0.105
bge-small	0.376	0.483	-0.108	-0.052
nomic	0.361	0.473	-0.112	-0.058

Table 19. Held-out skill of each learned fit on each model's $k = 20$ papers graph.

learned fit	scored on minilm	scored on bge-small	scored on nomic
fitted on minilm	0.922	0.901	0.891
fitted on bge-small	0.913	0.938	0.909
fitted on nomic	0.899	0.912	0.918

5.12 The default scheme under a stated rule

The tool needs one default scheme for a general library of mostly books, filed by its classifier alone. The decision record (`docs/decisions/default-scheme.md`) states the rule: among placed runs whose coverage is at least a floor, take the highest skill; every run whose paired difference to it has a 95% interval covering zero is tied; the tied run with the lowest navigation cost wins. The floor is 0.90, so that at most one book in ten waits for a person. The rule was applied in each of the 27 book settings of Section 5.10. Table 20 gives the winners by floor, over all

placed schemes and over the open schemes the tool may bundle (FDC, the LCC outline with and without LCSH vocabulary, LCC full, BK and the learned scheme).

Table 20. Winner of the default-scheme rule on the 3,000 books, settings won out of 27, by coverage floor.

coverage floor	winner, all placed schemes (settings)	winner, bundleable open schemes (settings)
0.70	`lcc-lcsh-placed` 15, `unesco-thesaurus-placed` 12	`lcc-lcsh-placed` 18, `fdc-placed` 9
0.80	`lcc-lcsh-placed` 15, `unesco-thesaurus-placed` 12	`lcc-lcsh-placed` 18, `fdc-placed` 9
0.85	`lcc-lcsh-placed` 15, `unesco-thesaurus-placed` 12	`lcc-lcsh-placed` 27
0.90	`lcc-lcsh-placed` 15, `unesco-thesaurus-placed` 12	`lcc-lcsh-placed` 27
0.95	`lcc-lcsh-placed` 27	`lcc-lcsh-placed` 27

At the 0.90 floor, `<!-- v:x_def_winner -->lcc-lcsh-placed<!-- /v -->`, the LCC outline with LCSH headings as search vocabulary, wins all 27 of 27 settings among the open schemes, and it is the only winner at every floor from 0.85 up. Over all placed schemes it wins 15 and the UNESCO Thesaurus 12; the thesaurus is not among the schemes the tool may bundle. Part of the win is vocabulary, not scheme: LCSH lifts the outline's coverage from 0.915 to 0.985, while FDC places 0.839 with its captions alone and falls below the floor. The bundled default of the tool is this scheme (`lcc@2024`). The decision record also documents an alternative for a catalogue-first pipeline, FDC cut to Dewey sections, which rests on the gold-number tie of Section 5.10 and on openness and notation length rather than on a metric result.

6. Discussion

What the measurement supports. The method gives one scale for schemes of different size, depth and domain. It costs little: no scoring run took more than 6.9 s, and the reference tree took 58 s on the corpus graph. Its inputs are pinned by hash. On gold labels it separates a well-fitted scheme (arXiv on arXiv) from a general scheme on books (LCC on the holdout) and from schemes with too little gold to judge (MSC, JEL). Within one classifier, the order of schemes on the papers does not depend on the embedding model, k or graph threshold (Section 5.9). Several book-side comparisons are stable in every setting we tried: the cut depth of a Dewey-derived scheme, and the cost of the full number against its three-digit section (Section 5.10).

What it does not support. It does not support a ranking of universal schemes, for four reasons that each have a number behind them.

1. *Coverage.* Classifier placement is the only way to reach most of the schemes, and coverage explains 59% of the rank variance of cost over the placed runs (Section 5.8). Gold placement does not escape this: MSC's and JEL's gold scores measure how few papers carry a code.
2. *The classifier.* The order of schemes changes when the classifier does: the Spearman correlation between the lexical and shortlist tiers' skill is 0.14, and the leader changes from openalex-topics to arxiv (Sections 5.8 and 5.9). The stopword fix shows the same risk from the other side: a classifier defect can change placements drastically while skill barely moves, or moves the wrong way.
3. *The objective.* FDC ranks above LCC under triplet accuracy in 27 of 54 settings and under cost in 0. Dewey sections against LCC on books is a setting-dependent tie (13 of 27 settings match the claimed sign under cost, 8 the tie under the corrected objective).
4. *The metric's validity.* The cost agrees with a blind judge at $\kappa = 0.27$. The false-join correction raises that to 0.48 against the judge and not against citation or subject-heading signals, and the judge is one model (Sections 5.6 and 5.7).

A scheme's rank on this metric is therefore a statement about a similarity graph, a classifier and an objective. It is not a statement about the scheme's quality in use. The benchmark's own consumers should read it that way. The default scheme follows a stated rule rather than a ranking: the bundled LCC outline with LCSH vocabulary wins it in 27 of 27 book settings among the open schemes (Section 5.12), and that win is partly a vocabulary effect. On librarian-assigned numbers the metrics do not separate Dewey from LCC, so a catalogue-first pipeline could choose either on other grounds (`docs/decisions/default-scheme.md`).

The learned scheme. A tree fitted to the objective sets the floor for how low the cost can go on a given graph. Its lead over human schemes is real in the sense that it holds on graphs it was not fitted to, and partly home advantage: it is larger on the fit graph (Section 5.11). It cannot be a scheme in the library sense without a notation and a way to file a new item that does not need the embedding model.

Toward a fair comparison. The benchmark design lays out controls (research/benchmark/DESIGN.md §2.5 and §3). A coverage control and a second classifier tier, an embedding, k and threshold sweep, item bootstrap intervals and a judge-based validity check have now been run. A permutation null that keeps each scheme's shape and class-size histogram, a label-intrinsic track that needs no works, a noise-injection curve that calibrates classifier error on LCC, and a placement by a classifier other than shelfmark's (Annif or a prompted LLM) have not. The largest remaining gain would come from a second annotator model and a human annotator on the validity sample, and from a graph that is not built from embeddings (co-use or citations).

Interpretation of skill. Skill compares a scheme with a tree fitted to the same similarity graph. A scheme encodes many relations besides textual similarity: discipline, form, audience, place, period. A low skill can mean the scheme organizes by something the embedding does not see. Our score measures agreement with one notion of relatedness. It does not measure quality in every sense that matters to libraries.

7. Limitations

- Classifier bias. Every non-gold placement comes from shelfmark's own classifier: the lexical profile for the main tables, and a BGE-small shortlist profile for Section 5.8. The lexical profile matches class labels and attached vocabulary, so a scheme with rich vocabulary is reached better than one with bare labels, independent of its structure. Most placements are below the classifier's own confidence target (Table 7). Table 8 and Section 5.8 show the classifier effect is as large as the effects we want to measure. No classifier outside this project (for example Annif or a prompted LLM) was run for comparison. The semantic tier's directory is marked pending in the results index, so those numbers are preliminary.
- One annotator for validity. The validity check used one LLM judge, on 953 pairs from 8 runs in three strata, so its shares describe the sample and not the population of pairs. The judge is self-consistent ($\kappa = 0.92$), but self-consistency is not validity. There is no human annotator and no second model. The non-LLM signals (subject headings, citations) are partial, and the citation signal is thin for recent preprints.
- Science-heavy corpus. 86.4% of the corpus is computer science, mathematics or physics. Humanities, law, arts and general nonfiction appear only in the 500-record holdout, which is itself 76.4% class P, and in the 3,000-book sample, whose graph text is mostly LC subject headings. Domain schemes (arXiv, MSC, ACM CCS) are scored on their home ground and general schemes are not, so no cross-type comparison is fair.
- Low gold coverage for MSC and JEL. Only 13.2% of papers carry a valid MSC code and 0.02% a JEL code. Unplaced items hang under the root, so these gold scores measure coverage, not structure. The covered-subset analysis (Section 5.8) is the control for this, and its subset scores use a restricted graph.
- ACM CCS gold missing, and no official crosswalk. The corpus field meant to carry ACM CCS labels holds the ACM classes submitted to arXiv, which are CCS 1998 codes. No value is a CCS 2012 concept path, so the gold run was skipped. ACM publishes no CCS 1998 to CCS 2012 mapping: it says it built one for its own digital library and exposes it only as terms on each article's page, and the published CCS 2012 SKOS file holds no mapping or 1998 codes (research/reports/5). We did not build a mapping, because any we made would be ours and not ACM's, so the ACM CCS placed run has no gold counterpart.
- Scheme licensing and the DDC position. Several schemes cannot be redistributed: ACM CCS (educational and research use only), JEL (no open licence), MSC (non-commercial, share-alike), EuroVoc (terms stated inconsistently) and ILC2 (conflicting licence statements) (Table 2). Their files are built locally from the publishers' downloads and verified by hash, but they are not published. Exact reproduction therefore de-

pends on the upstream downloads staying available and unchanged. DDC, which OCLC describes as the world's most widely used way to organize library collections, is owned by OCLC (OCLC, n.d.). We use Dewey class *numbers* from public catalogue records, and the tree is the notation's digit prefixes. No Dewey caption, schedule, table, Manual entry or relative-index term from any edition is used, and nothing is taken from OCLC services. The position rests on class numbers being short numeric strings in public catalogue data; it is an engineering position, not legal advice, and it is written for the United States and the European Union only (research/reports/7). The Dewey-derived scheme that can be bundled and used to *place* items is FDC (Ockerbloom, n.d.). Even LCC's open status depends on jurisdiction. The Library of Congress calls its linked data a public-domain data set (Library of Congress 2026a), but its Classification Web notice says LC-originated classification data are copyrighted except within the USA (Library of Congress, n.d.).

- Titles-only holdout. The book holdout has no descriptions. Its similarity graph is built from titles averaging 3.5 words, and the classifier finds candidates for few of them. The holdout is also small ($n = 500$). Tables 3 to 10 give point values; intervals for the placed runs are in the bootstrap runs of `bench/results/2026-10-10-covered` and `-semantic`.
- Method limits. The kNN-sparsified cost agrees with a dense cost on random subsets (Section 5.8) but has not been checked on the full corpus. The embedding-model, k and threshold sweep uses three models; the similarity is always an embedding. Polyhierarchies are reduced to one parent per class. Only one placement per item is used, although 35.5% of a 200-work Open Library sample had more than one LCC value (DESIGN.md §0). The balance statistics count each item at every ancestor class (Section 3.5). The navigation models are unfitted. The 3,000-book sample is stratified equally over LCC's main classes, which favours LCC's false-join term.
- Search for prior work. The novelty claim rests on the search in research/reports/8-prior-work-search.md: 94 Crossref queries over the JASIS/JASIST family, Knowledge Organization, Cataloging & Classification Quarterly, ISKO, SIG/CR and NASKO proceedings, JCDL and TPDL/ECDL, plus five web searches. It screened titles and abstracts only. Full texts and non-English literature were not searched, one person screened, and Crossref's coverage of proceedings is incomplete (ISKO proceedings before 2012 are missing).

8. Reproducibility statement

All inputs and outputs are in the repository or pinned by hash.

- Corpus. `bench/corpus/corpus.jsonl` and `holdout.jsonl`, with ID lists and SHA-256 hashes (bench/corpus/README.md). Metadata only, CC0 1.0. The harvest scripts and the deterministic Rust sampler are in `bench/corpus/builder/`.
- Similarity graphs. Published gzipped, with manifests, in `bench/results/2026-10-10/graphs/`. The manifests give the model revision `d13954661f83248295ba75c1ed411eef3b7b936e`, the `model.onnx` SHA-256, τ, k and the graph SHA-256 (corpus graph 759d3b8461a9384d…).
- Harness. `crates/bench`, at the commit recorded in `results.json` (092494850adc). The classifier for the placed run is at f4ce24b78eda (`profile:lexical@2`). Both binaries' SHA-256 hashes are recorded too.
- Schemes. Bundled schemes are in the repository. Fetch-only schemes are rebuilt by `crates/bench/scripts/build_scheme` from the publishers' downloads, checked against committed `sources.lock.json` files, and must match the `file_sha256` recorded in `results.json`.
- Commands. `bench/results/2026-10-10/REPRODUCE.md`, `bench/results/2026-10-10-r2/REPRODUCE.md` and `bench/results/2026-10-10-r3/REPRODUCE.md` list every command. The first run took about 45 minutes and the r2 rerun about 25 minutes, both on 8 CPUs.
- Later results. `docs/validity/` (sample, rubric, labels and tables), `bench/results/2026-10-10-covered`, `-embed`, `-semantic`, `-learned`, `2026-10-11-falsejoin`, `-stability` and `-book-stability`, each with a `REPRODUCE.md`. The judge's labels are in `docs/validity/sample.jsonl`; rerunning the judge needs the same model and rubric.
- Determinism. With the same graph and scheme files, gold scores reproduce exactly. Placed scores reproduce exactly with the same classifier build.
- This paper. `paper/paper1/scripts/make_tables.py` regenerates every table and marked number from the JSON, using `paper/common/extra_results.py` for the validity, stability, book and semantic-tier values. `--check` fails if `paper.md` is stale. It also checks that the r2 and r3 runs used the 2026-10-10 graphs

and scheme files, and that the corpus files match their recorded hashes.

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